

SINGLEMODE FIBRE TO ITU-T RECOMMENDATION G.652

The following details summarise the main Optical, Geometrical and Physical Characteristics of fibres, compliant to ITU-T Rec. G.652.

Fibres are manufactured from high grade silica, doped as necessary to achieve the required light guiding properties, designed with a matched-cladding, step-index profile.

The fibre coating is a dual layer structure of ultra-violet cured acrylate resin. The lower modulus inner layer being optimised for both adhesion to the fibre surface and mechanical stripping using the appropriate stripping tools. The outer layer is optimised for abrasion resistance and fibre processing properties.

ATTENUATION REQUIREMENTS

Wavelength	Attenuation	Units
1310 nm	0.38 max	dB/km
1550 nm	0.26 max	dB/km

OTHER REQUIREMENTS

Parameter	Units	Generic Value for all grades
Cut-Off Wavelength (cables)	nm	≤1260
PMD _c (IEC concatenated value)	ps /√ km	≤ 0.5
Core Concentricity	μm	≤ 0.74
Mode Field Diameter	μm	8.8 – 9.6
Clad Diameter	μm	125 +/- 1
Attenuation at water peak (1385 nm)	dB/km	≤1.5
Attenuation Uniformity (over 2km)	dB/km	≤0.05
Point Defects	dB	≤0.1
Attenuation with Bending	dB	≤0.05
Dispersion at 1285 – 1330nm	ps/(nm.km)	≤3.5
Dispersion at 1550nm	ps/(nm.km)	≤18.0
Zero Dispersion Wavelength	nm	1300 – 1324
Dispersion Slope	ps/(nm ² .km)	≤0.092
Cladding non-circularity	%	≤1.5
Coating Diameter – Coloured	μm	250 +/-15
Coating concentricity	μm	≤10.0
Strip Force	N	1 – 3.5
Proof Test	%	≥1
Stress Corrosion (Static Fatigue)		≥23

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